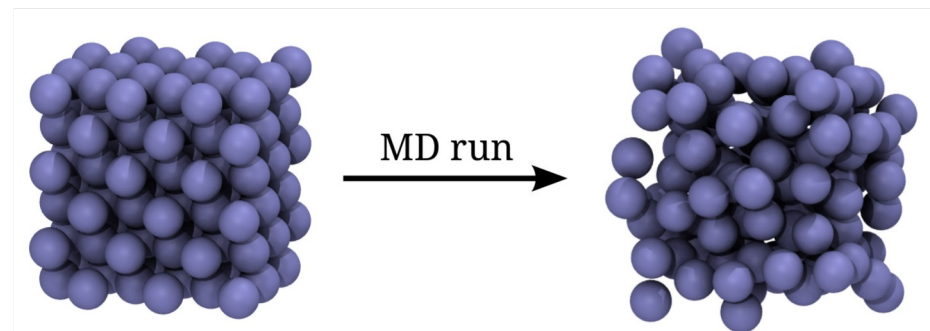


Aula 01 – Introdução ao Python



Prof. Elvis Soares
elvis@peq.coppe.ufrj.br

Infos

Horário de aulas: 3a, 09:00-12:00

Sala de Aula: I224

Calendário: 10/Mar – 07/Jul (~17 encontros)

Professores: Frederico Tavares (tavares@eq.ufrj.br)

Elvis Soares (elvis@peq.coppe.ufrj.br)



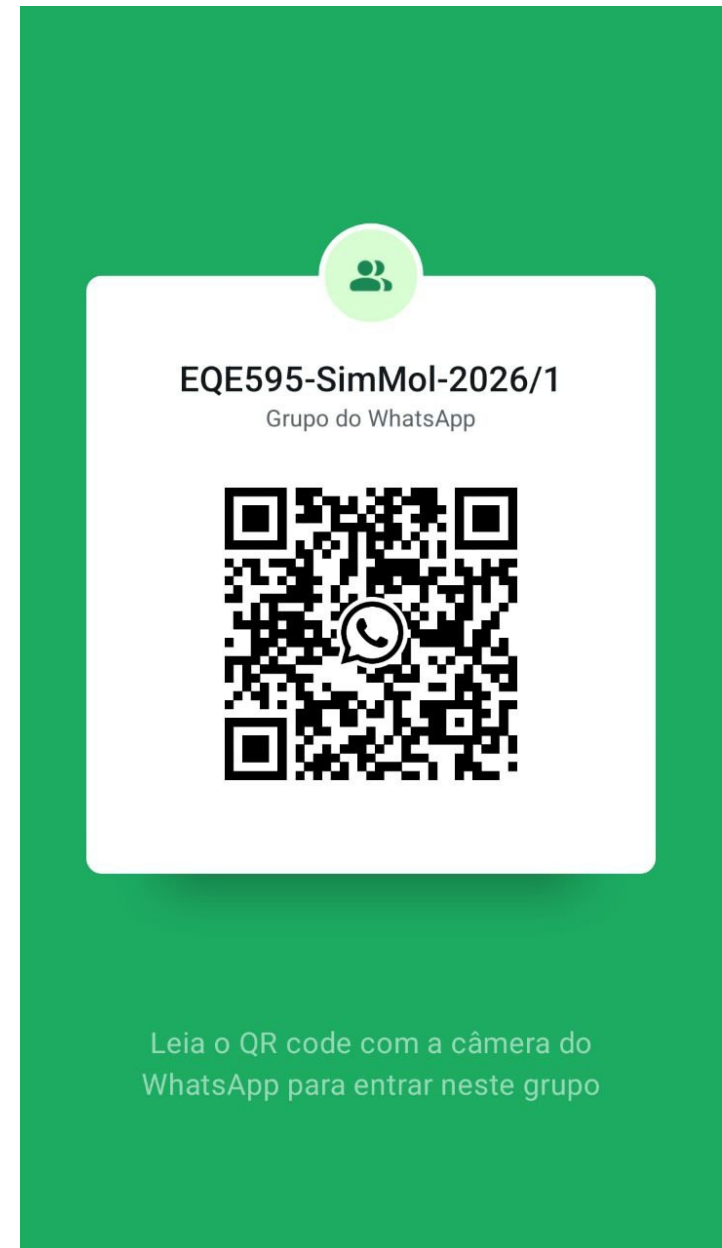
<https://github.com/elvissoares/EQE595-SimMol>

CRITÉRIOS DE AVALIAÇÃO

- 20% de Presença e Participação
- 30% de Listas de Exercícios (~06 listas)
- 50% de Projeto Final

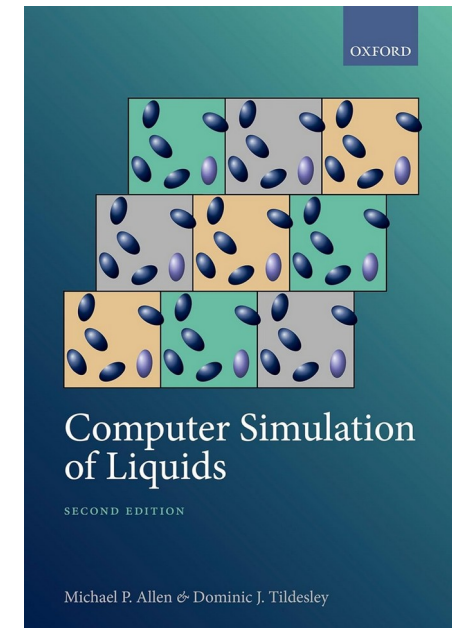
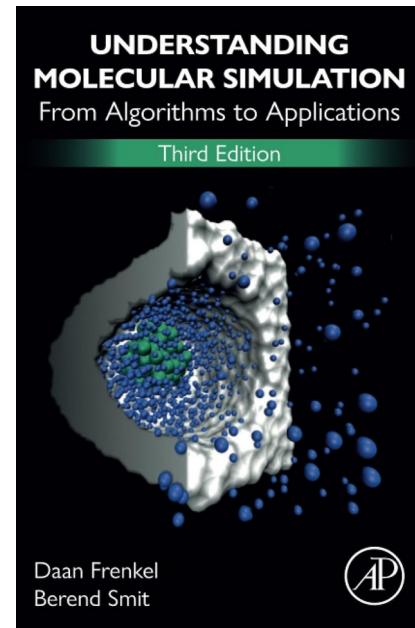
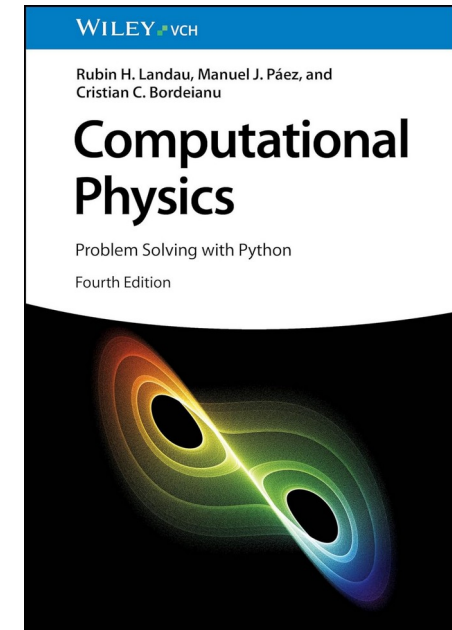
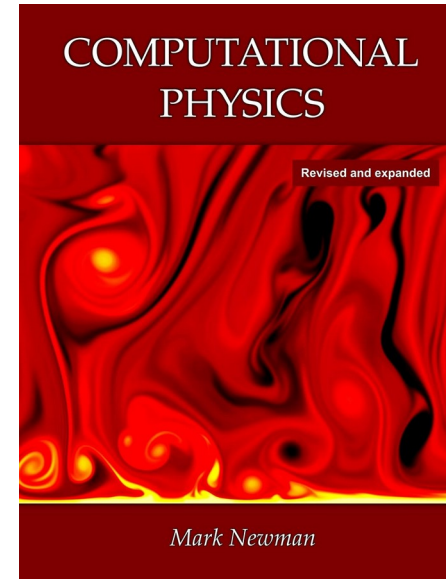
Ementa

1. Introdução ao Python
2. Interação de Lennard-Jones
3. Ensemble NVE e integrador de Verlet
4. Ensemble NVT e integrador de Langevin
5. Ensemble NPT
6. Água e Campos de Força
7. Solute e Solvente
8. Integração Termodinâmica



Referências

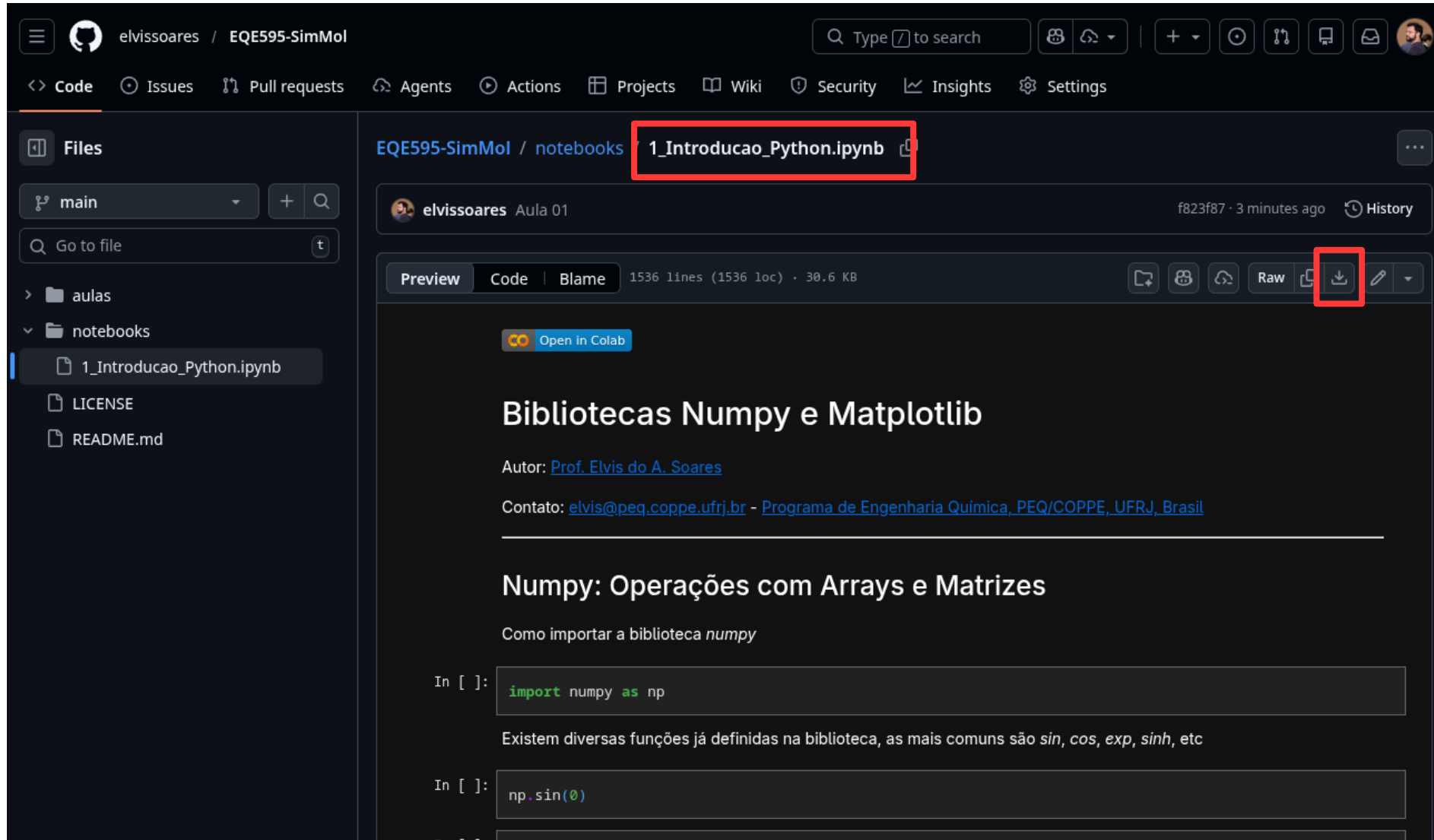
- Newman, M. (2013). **Computational physics**. CreateSpace Independent Publishing Platform.
- Landau, R. H., Páez, M. J., & Bordeianu, C. C. (2024). **Computational physics: Problem solving with Python, 4th Edition**. John Wiley & Sons.
- Frenkel, D., & Smit, B. (2023). **Understanding molecular simulation: from algorithms to applications, 3rd Edition**. Elsevier.
- Allen, M. P., & Tildesley, D. J. (2017). **Computer simulation of liquids, 2nd Edition**. Oxford university press.



Ferramentas Computacionais



Baixando o Notebook



The screenshot shows the GitHub interface for the repository `elvissoares / EQE595-SimMol`. The `Code` tab is selected. In the `Files` sidebar, the `notebooks` folder is expanded, and the file `1_Introducao_Python.ipynb` is highlighted. The main content area shows the notebook's preview. The file name `1_Introducao_Python.ipynb` is highlighted with a red box. In the top right of the notebook preview, the download icon (a square with a downward arrow) is also highlighted with a red box. The notebook content includes a title `Bibliotecas Numpy e Matplotlib`, author information, and code cells for importing `numpy` and using `np.sin()`.

elvissoares / EQE595-SimMol

Code Issues Pull requests Agents Actions Projects Wiki Security Insights Settings

Files

main

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notebooks

1_Introducao_Python.ipynb

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README.md

EQE595-SimMol / notebooks 1_Introducao_Python.ipynb

elvissoares Aula 01 f823f87 · 3 minutes ago History

Preview Code Blame 1536 lines (1536 loc) · 30.6 KB

Open in Colab

Bibliotecas Numpy e Matplotlib

Autor: [Prof. Elvis do A. Soares](#)

Contato: elvis@peq.coppe.ufrj.br - Programa de Engenharia Química, PEQ/COPPE, UFRJ, Brasil

Numpy: Operações com Arrays e Matrizes

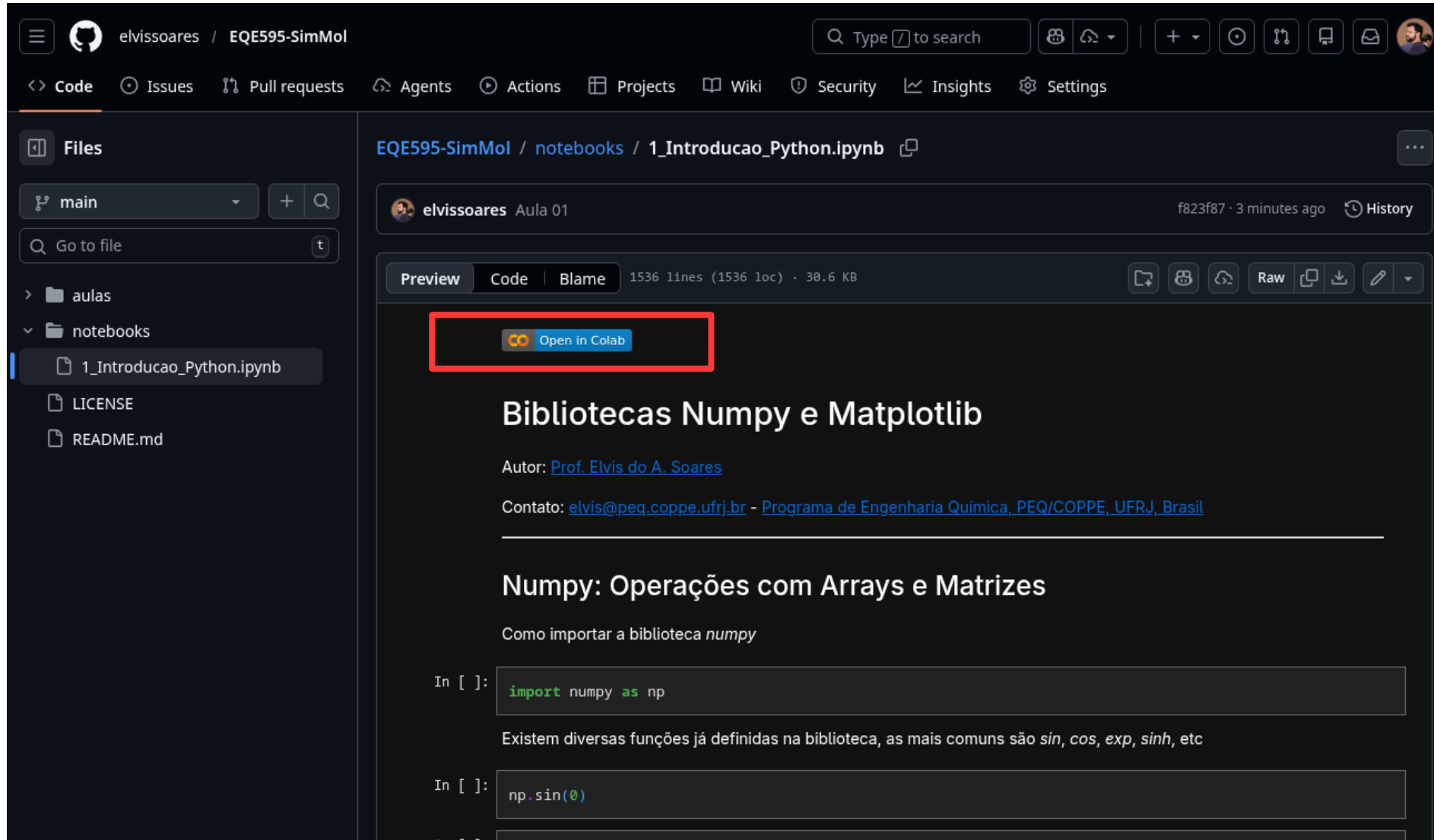
Como importar a biblioteca `numpy`

```
In [ ]: import numpy as np
```

Existem diversas funções já definidas na biblioteca, as mais comuns são `sin`, `cos`, `exp`, `sinh`, etc

```
In [ ]: np.sin(0)
```

Abrindo o Notebook no Colab



The screenshot shows the GitHub interface for the repository `elvissoares / EQE595-SimMol`. The left sidebar displays the file structure with folders `aulas` and `notebooks`, and files `1_Introducao_Python.ipynb`, `LICENSE`, and `README.md`. The main content area shows the notebook `1_Introducao_Python.ipynb` in the `Preview` tab. A red rectangle highlights the `Open in Colab` button, which is located above the notebook's title `Bibliotecas Numpy e Matplotlib`. Below the title, the author is listed as `Prof. Elvis do A. Soares` and the contact information is `elvis@peq.coppe.ufrj.br - Programa de Engenharia Química, PEQ/COPPE, UFRJ, Brasil`. The notebook content includes a section titled `Numpy: Operações com Arrays e Matrizes` and a code cell showing the import of `numpy` as `np`.

Files

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notebooks

1_Introducao_Python.ipynb

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EQE595-SimMol / notebooks / 1_Introducao_Python.ipynb

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Bibliotecas Numpy e Matplotlib

Autor: [Prof. Elvis do A. Soares](#)

Contato: elvis@peq.coppe.ufrj.br - Programa de Engenharia Química, PEQ/COPPE, UFRJ, Brasil

Numpy: Operações com Arrays e Matrizes

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```

Usando Python pelo Colab

The screenshot displays the Google Colab interface for a notebook titled "1_Introducao_Python.ipynb". The top navigation bar includes the Colab logo, a menu (File, Edit, View, Insert, Runtime, Tools, Help), a settings gear, a "Share" button, and a user profile picture. Below this is a toolbar with "Commands", "Code", "Text", "Run all", and "Copy to Drive" options. A left sidebar contains icons for file management and viewing. The main content area shows a document with the title "Bibliotecas Numpy e Matplotlib". Below the title, it lists the author as "Prof. Elvis do A. Soares" and provides contact information: "elvis@peq.coppe.ufrj.br" and "Programa de Engenharia Química, PEQ/COPPE, UFRJ, Brasil". The document then transitions into a code section titled "Numpy: Operações com Arrays e Matrizes", which includes the instruction "Como importar a biblioteca *numpy*" and a code cell containing `import numpy as np`. Below the code cell, it states "Existem diversas funções já definidas na biblioteca, as mais comuns são *sin*, *cos*, *exp*, *sinh*, etc". At the bottom of the interface, there are tabs for "Variables" and "Terminal", and a blue circular button with a star icon.

1_Introducao_Python.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all Copy to Drive

Connect

Bibliotecas Numpy e Matplotlib

Autor: [Prof. Elvis do A. Soares](#)

Contato: elvis@peq.coppe.ufrj.br - [Programa de Engenharia Química, PEQ/COPPE, UFRJ, Brasil](#)

Numpy: Operações com Arrays e Matrizes

Como importar a biblioteca *numpy*

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Variables Terminal